

Phase 1 – Data Pre-processing Using Excel

1. Data Cleaning in Cleaned_Data Sheet

- Removed duplicate records from the raw Nike sales dataset.
- Cleaned the **Size** column by replacing blank values with NA and standardizing valid text values using UPPER().
- Handled missing values in the **Units_Sold** column using the median of valid positive values.
- Handled missing and invalid values in the **MRP** column using the median of valid positive MRP values.
- Cleaned the **Discount_Applied** column by replacing blank values with 0.
- Standardized the **Region** column using Find and Replace. Variations such as Hyd, hyderabad, and Hyderabad were standardized to **Hyderabad**, while bengaluru and bangalore was standardized to **Bangalore**.
- Prepared the cleaned dataset for further analysis.

2. PivotTable Analysis

Created the following PivotTables in Excel:

- **Sales by Product Line**
- **Sales by Region**
- **Sales by Gender**
- **Sales by Year and Month**

These PivotTables were created to summarize the cleaned data and understand the major sales patterns before moving to Power BI.

3. Fact and Dimension Tables

Created a structured **Fact and Dimension table model** in Excel for the next phase of the project.

Fact Table:

- Fact_Sales

Dimension Tables:

- Dim_Product

- Dim_Region
- Dim_SalesChannel
- Dim_Gender
- Dim_Date

4. Dimension Table Preparation

- **Dim_Product:** Created Product_Key, Product_Name, Product_Line, Size, and Product_Lookup_Key.
- **Dim_Region:** Created Region_Key and standardized Region.
- **Dim_SalesChannel:** Created Channel_Key and Sales_Channel.
- **Dim_Gender:** Created Gender_Key and Gender_Category.
- **Dim_Date:** Created Date_Key, Date, Year, Quarter, Month_Number, Month, and Month_Year.

5. Final Outcome

The raw Nike sales dataset was cleaned and transformed in Excel, summarized using PivotTables, and organized into Fact and Dimension tables. The prepared Excel model will be used as the foundation for **Phase 2 – Power BI data modeling, visualization, and dashboard development.**